Manifold-Aware Concept Erasure and Evaluation

Anonymous Author(s)
Affiliation
Address
email

Abstract

Concept erasure methods that operate in the ambient representation space can push representations off the data manifold, creating spurious correlations rather than removing information. We argue that both erasure and its evaluation should respect the intrinsic geometry of the representation manifold. On the erasure side, we introduce Tangent Gradient-based Concept Erasure (TanGCE), which constrains erasure gradients to the local tangent space estimated via neighborhood PCA. On the evaluation side, we adopt a k-nearest-neighbor mutual information estimator that measures concept information through local neighborhood structure—the same geometric lens that TanGCE uses for intervention. We complement this with unsupervised sparse autoencoder diagnostics that detect feature-level redistribution without requiring control-task labels. Across three concept domains—sycophancy, gender, and safety—in Qwen2.5-0.5B, TanGCE achieves the lowest nonlinear leakage while preserving control-task mutual information and maintaining near-zero feature redistribution. A λ sweep reveals that ambient erasure at high step sizes *increases* concept mutual information by laundering it into off-manifold displacement, a failure mode that TanGCE avoids by construction.

1 Introduction

Effective concept erasure in large language models requires a delicate balance between removing targeted information and preserving the model's broader utility. While gradient-based methods offer a path toward targeted intervention, they often suffer from substantial performance degradation. A natural hypothesis is that this collapse is driven in part by off-manifold edits: erasure directions computed in the high-dimensional ambient space can point orthogonal to the data manifold, displacing representations into low-density regions where the model's computation becomes unstable.

To address this failure mode, we propose Gradient-based Concept Erasure with Tangent projection (`gce_tangent`). By constraining erasure gradients to the local tangent space of the representation manifold, `gce_tangent` aims to remove concept-specific structure while preserving more of the underlying distributional geometry. This approach contrasts with `gce_ambient`, which applies raw gradients in the full hidden space and therefore provides a direct baseline for the value of tangent projection.

Our contributions are threefold:

1. **TanGCE: manifold-aware concept erasure.** We introduce Tangent Gradient-based Concept Erasure (TanGCE), which constrains erasure gradients to the local tangent space of the representation manifold. By retaining the tangent-projected gradient norm as a per-sample confidence weight, TanGCE naturally attenuates erasure for samples whose concept signal is predominantly off-manifold. We demonstrate that TanGCE achieves the lowest nonlinear leakage across three concept domains—sycophancy, gender, and safety—while preserving downstream task performance.
2. **Manifold-aware mutual information evaluation.** We adopt a k-nearest-neighbor MI estimator that measures concept information through local neighborhood structure on the

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

representation manifold. The estimator computes $I(\mathbf{X}; y)$ by examining, for each point, how many cross-class neighbors fall within the distance to its k-th same-class neighbor—a purely geometric quantity that captures all information (linear and nonlinear) without assuming a probe architecture. This creates a coherent geometric framework: TanGCE uses local PCA neighborhoods to estimate tangent spaces for erasure, while the k-NN MI estimator uses local ball neighborhoods to estimate information content. Both operate through local neighborhood structure rather than global ambient-space assumptions, and both degrade gracefully when the manifold assumption is only approximately satisfied. We compute MI for both concept labels (should decrease after erasure) and control labels (should remain unchanged), yielding a complete picture of erasure selectivity.

3. **Unsupervised SAE-based erasure diagnostics.** We propose three metrics computed via sparse autoencoders trained on a general-purpose corpus (no concept labels needed): *concept feature removal rate* (fraction of concept-correlated SAE features disrupted), *collateral damage rate* (fraction of non-concept features disrupted), and *feature redistribution score* (fraction of previously non-concept features that become concept-correlated after erasure). These metrics measure the surgicality and precision of erasure without requiring annotated control-task labels, and reveal failure modes invisible to downstream probes—including the critical finding that ambient erasure at high λ *creates* new concept correlations through off-manifold displacement (Section 5.4).

We evaluate on sycophancy as the primary nonlinear concept, gender (Bias in Bios) as a diagnostic baseline, and safety as a secondary case study. Using Qwen2.5-0.5B at layer 8, we show that TanGCE achieves the lowest nonlinear leakage across all three domains while maintaining near-zero feature redistribution and preserving control-task mutual information.

2 Notation and Formal Framework

We define the notation used throughout, then give the formal definitions of TanGCE, the k-NN MI estimator, and the SAE-based diagnostics.

Language model and encoder. Let Σ^* denote the set of all strings over a vocabulary Σ. A language model p defines a distribution over Σ^*. We write $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ for the hidden representation at layer ℓ, where d is the hidden dimension. For a fixed layer ℓ (omitted when clear from context), the representation of string s is $\mathbf{x} = \mathrm{enc}_\ell(s) \in \mathbb{R}^d$.

Concepts and data. A binary concept is defined by a labeling function $\phi : \Sigma^* \to \mathcal{C}$, where $\mathcal{C} = \{0, 1\}$. Each sample i has representation $\mathbf{x}_i \in \mathbb{R}^d$ and concept label $y_i = \phi(s_i) \in \mathcal{C}$, with an optional control label c_i (e.g., profession for gender erasure). We write $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ for the dataset and $\mathcal{D}_c = \{i : y_i = c\}$ for the class-c subset, with $N_c = |\mathcal{D}_c|$.

The class-conditional representation is the random variable $X_c \sim \mathrm{enc}_\ell(s) \mid \phi(s) = c$, i.e., the distribution of hidden states for strings labeled c.

Concept scorer. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a two-layer MLP trained with binary cross-entropy on $\mathcal{D}$ to predict y from $\mathbf{x}$. Its per-sample gradient with respect to the input is $\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i) \in \mathbb{R}^d$, which points in the direction of steepest increase in concept probability at $\mathbf{x}_i$.

Conventions. Index i ranges over samples. $\|\cdot\|$ denotes the Euclidean norm unless stated otherwise. ψ denotes the digamma function $\psi(x) = \frac{d}{dx} \ln \Gamma(x)$. $\langle \cdot \rangle$ denotes the sample average over all N points.

2.1 Representation Surgery Baselines

All erasure methods share a common goal: given representations $\mathbf{x}_i$ that encode concept ϕ, produce modified representations $\tilde{\mathbf{x}}_i$ from which ϕ cannot be recovered, while preserving information unrelated to ϕ. We review the three linear baselines before introducing the nonlinear GCE family.

Linear projection. Given a concept direction $\mathbf{w} \in \mathbb{R}^d$ (typically the difference of class-conditional means, $\mathbf{w} = \mathbb{E}[X_1] - \mathbb{E}[X_0]$), linear projection removes the component along $\mathbf{w}$:

$$\tilde{\mathbf{x}}_i = (\mathbf{I} - \hat{\mathbf{w}}\hat{\mathbf{w}}^\top)\mathbf{x}_i, \qquad \hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|. \tag{1}$$

This removes a single linear direction but leaves all other concept-correlated structure intact.

INLP (Iterative Nullspace Projection). INLP (Ravfogel et al., 2020) iteratively trains linear classifiers $\mathbf{w}_1, \ldots, \mathbf{w}_T$ and projects onto the intersection of their nullspaces:

$$\mathbf{P}_{\text{INLP}} = \prod_{t=1}^{T} (\mathbf{I} - \hat{\mathbf{w}}_t \hat{\mathbf{w}}_t^\top), \qquad \tilde{\mathbf{x}}_i = \mathbf{P}_{\text{INLP}}\, \mathbf{x}_i. \tag{2}$$

Each $\mathbf{w}_t$ is trained on the residual after projecting out $\mathbf{w}_1, \ldots, \mathbf{w}_{t-1}$. INLP extends linear projection to multiple directions but remains limited to linearly decodable concept information.

LEACE (Least-squares Concept Erasure). LEACE (Belrose et al., 2023) finds an oblique projection $\mathbf{P} \in \mathbb{R}^{d \times d}$ of rank $|\mathcal{C}| - 1$ and a translation $\mathbf{b}$ that minimally modify representations while ensuring no log-linear classifier can recover the concept:

$$\tilde{\mathbf{x}}_i = \mathbf{P}\mathbf{x}_i + \mathbf{b}. \tag{3}$$

LEACE guarantees $\mathbb{E}[\tilde{X}_c] = \mathbb{E}[X]$ and $\mathrm{Var}[\tilde{X}_c] = \mathrm{Var}[\tilde{X}_{c'}]$ for all $c, c' \in \mathcal{C}$, erasing both mean and covariance differences between class-conditional distributions. This provides the strongest linear erasure guarantee among the baselines.

2.2 Gradient-Based Concept Erasure

The linear baselines above remove linearly decodable concept information but leave nonlinear structure intact. GCE addresses this by computing per-sample erasure directions from the gradient of a nonlinear concept scorer.

Ambient GCE. Given a trained scorer f_θ, the ambient erasure direction at sample i is the unit gradient:

$$\mathbf{u}_i = \frac{\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)}{\|\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)\|}. \tag{4}$$

The erased representation is obtained by projecting out the concept component:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\, (\mathbf{x}_i^\top \mathbf{u}_i)\, \mathbf{u}_i, \tag{5}$$

where $\lambda > 0$ controls the erasure strength. At $\lambda = 1$, this removes the full projection onto $\mathbf{u}_i$; at $\lambda < 1$, it partially attenuates.

Local tangent space estimation. For each query point $\mathbf{x}_i$, we estimate the tangent space $T_{\mathbf{x}_i}\mathcal{M}$ of the data manifold $\mathcal{M}$ via local PCA on its k-nearest neighbors. Let $\mathcal{N}_k(\mathbf{x}_i) = \{\mathbf{x}_{j_1}, \ldots, \mathbf{x}_{j_k}\}$ denote the k nearest neighbors of $\mathbf{x}_i$ in the training set. We form secant vectors $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ and compute the local covariance:

$$\mathbf{C}_i = \frac{1}{k} \sum_{m=1}^{k} \mathbf{s}_{j_m} \mathbf{s}_{j_m}^\top \in \mathbb{R}^{d \times d}. \tag{6}$$

The top r eigenvectors of $\mathbf{C}_i$ form the tangent basis $\mathbf{B}_i = [\mathbf{b}_i^{(1)}, \ldots, \mathbf{b}_i^{(r)}] \in \mathbb{R}^{d \times r}$, where the intrinsic dimension r is determined automatically by retaining eigenvectors that explain a cumulative fraction of the variance. The orthogonal projection onto $T_{\mathbf{x}_i}\mathcal{M}$ is:

$$\mathbf{P}_{T_i} = \mathbf{B}_i \mathbf{B}_i^\top. \tag{7}$$

TanGCE. TanGCE projects the ambient erasure direction onto the local tangent space:

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\,\mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i). \tag{8}$$

The projected direction $\mathbf{u}_i^{\tan}$ is *not* re-normalized to unit length. Its norm $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as a per-sample confidence weight: when the concept gradient is mostly off-manifold, $\|\mathbf{u}_i^{\tan}\| \ll 1$, and the effective erasure step is attenuated. The erased representation is:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\,(\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\,\mathbf{u}_i^{\tan}. \tag{9}$$

The effective step size is $\lambda\|\mathbf{u}_i^{\tan}\|^2$, which is maximized when the gradient lies entirely in the tangent space and vanishes when it is entirely off-manifold.

Why tangent projection helps. Ambient GCE (Eq. 5) displaces $\mathbf{x}_i$ along $\mathbf{u}_i$, which may point off the data manifold. At high λ, this pushes representations into low-density regions where the displacement magnitude itself correlates with concept signal strength, creating spurious concept information (Section 5.4). TanGCE constrains the displacement to the tangent space, ensuring that the modified representation stays on (or near) $\mathcal{M}$.

2.3 Mutual Information Evaluation

We estimate $I(\mathbf{X}; y)$ using the Ross/Gao k-nearest-neighbor estimator for mutual information between a continuous variable $\mathbf{X} \in \mathbb{R}^d$ and a discrete variable y.

Procedure. For each sample i:

1. Find the k-th nearest neighbor of $\mathbf{x}_i$ *within its class* $c(i) = y_i$. Let ε_i be the Euclidean distance to this neighbor.
2. Count m_i: the number of points from *all classes* within distance ε_i of $\mathbf{x}_i$ (excluding $\mathbf{x}_i$ itself).

The MI estimate is:

$$\hat{I}(\mathbf{X}; y) = \psi(k) - \big\langle \psi(m_i) \big\rangle + \psi(N) - \big\langle \psi(N_{c(i)}) \big\rangle, \tag{10}$$

where ψ is the digamma function, $N_{c(i)}$ is the number of samples in the class of point i, and $\langle \cdot \rangle$ averages over all N samples.

Interpretation. The estimator measures concept information through local neighborhood structure. The term $\psi(m_i) - \psi(k)$ captures the local class purity: when same-class points cluster ($m_i \approx k$, mostly same-class neighbors within ε_i), this term is small and MI is high. When classes are geometrically mixed ($m_i \gg k$, many cross-class neighbors within ε_i), this term is large and MI drops. The terms $\psi(N) - \psi(N_{c(i)})$ correct for class imbalance.

Erasure evaluation. We compute MI on both clean and erased representations:

- $\hat{I}(\mathbf{X}; y) - \hat{I}(\tilde{\mathbf{X}}; y) > 0$: concept information was removed.
- $\hat{I}(\mathbf{X}; c) \approx \hat{I}(\tilde{\mathbf{X}}; c)$: control-task information was preserved.

A method that reduces concept MI while preserving control MI achieves selective erasure.

2.4 Unsupervised SAE Diagnostics

We train a TopK sparse autoencoder (SAE) on a general-purpose corpus (WikiText), independent of any concept labels. The SAE maps representations to sparse feature activations: $\mathbf{z} = \mathrm{TopK}(\mathbf{W}_{\mathrm{enc}}(\mathbf{x} - \mathbf{b}_{\mathrm{dec}}) + \mathbf{b}_{\mathrm{enc}})$, where $\mathbf{z} \in \mathbb{R}^D$ with $D = \eta d$ (η is the expansion factor) and at most k entries are nonzero.

Feature–concept correlation. For each SAE feature $j \in \{1, \ldots, D\}$ active on the clean data ($\bar{z}_j > \tau_{\text{act}}$), we compute the point-biserial correlation between its activations and the concept label:

$$r_j = \frac{\bar{z}_j^{(1)} - \bar{z}_j^{(0)}}{\hat{\sigma}_{z_j}} \cdot \sqrt{\frac{N_0 N_1}{N^2}}, \tag{11}$$

where $\bar{z}_j^{(c)}$ is the mean activation of feature j for class c and $\hat{\sigma}_{z_j}$ is the pooled standard deviation. Features with $|r_j| > \tau_{\text{corr}}$ are classified as *concept features*; the remainder are *non-concept features*.

Disruption. A feature j is *disrupted* if its relative activation change exceeds a threshold:

$$\frac{|\bar{z}_j^{\text{erased}} - \bar{z}_j^{\text{clean}}|}{\bar{z}_j^{\text{clean}}} > \delta, \tag{12}$$

where $\bar{z}_j^{\text{clean}}$ and $\bar{z}_j^{\text{erased}}$ are mean activations on clean and erased data, and δ is the disruption threshold.

Metrics. Let $\mathcal{F}$ and $\mathcal{F}^c$ denote the concept and non-concept feature sets.

$$\text{Concept removal rate} = \frac{|\{j \in \mathcal{F} : j \text{ disrupted}\}|}{|\mathcal{F}|}, \tag{13}$$

$$\text{Damage rate} = \frac{|\{j \in \mathcal{F}^c : j \text{ disrupted}\}|}{|\mathcal{F}^c|}. \tag{14}$$

Feature redistribution. We recompute point-biserial correlations on the *erased* data. Features that were non-concept on clean data but become concept-correlated ($|r_j^{\text{erased}}| > \tau_{\text{corr}}$) on erased data indicate concept information *redistribution* rather than removal:

$$\text{Redistribution score} = \frac{|\{j \in \mathcal{F}^c : |r_j^{\text{erased}}| > \tau_{\text{corr}}\}|}{|\mathcal{F}^c|}. \tag{15}$$

A method with high removal, low damage, and zero redistribution achieves surgical concept erasure.

3 Implementation

We implement GCE using an iterative protocol with probe recycling. Each round of erasure begins by training a fresh concept scorer on the current (possibly already partially erased) representations, computing per-sample gradients, and applying the erasure step (Eq. 5 for Ambient GCE, Eq. 9 for TanGCE). The tangent estimator is refit in every round to track the evolving manifold geometry as representations shift.

Two diagnostics verify tangent quality throughout the iterative process. *Tangent Gradient Capture* (TGC) measures the fraction of the leakage gradient that lies within the estimated tangent space. *Tangent Neighbor Residual* (TNR) measures the fraction of local secant vectors not explained by the tangent basis. High TGC and low TNR confirm that the tangent estimation remains geometrically sound.

4 Experiments

Our experimental evaluation focuses on characterizing the performance of `gce_tangent` against `gce_ambient` across three distinct concept domains. We prioritize nonlinear concepts where manifold geometry is expected to play a critical role, while maintaining a diagnostic baseline for linear comparison.

4.1 Experimental Domains and Ordering

We evaluate our methods across three primary blocks, ordered by their relevance to nonlinear manifold erasure:

Table 1: Comparison of concept erasure methods by implementation status and role in this work. We distinguish between methods implemented in this repository and those positioned as literature baselines.

Method	Family	Implemented	Replication	Canonical Dataset	Role in Paper
Linear Projection	Linear	Yes	Yes	Bias in Bios	Global baseline
INLP	Linear	Yes	Yes	Bias in Bios	Linear baseline
LEACE	Linear	Yes	Yes	Bias in Bios / Syc.	Matched linear baseline
IGBP	Nonlinear	No	Not yet	Bias in Bios	Prior lineage (proxy only)
`gce_ambient`	Nonlinear	Yes	Yes	Sycophancy	Nonlinear baseline
`gce_tangent`	Nonlinear	Yes	Yes	Sycophancy	Main contribution

1. **Sycophancy (Primary Nonlinear Case):** Our central focus is on removing agreement bias in model responses. This represents a critical challenge for model alignment where concept encoding is expected to be locally curved.
2. **Safety (Secondary Case Study):** We evaluate the surgicality of erasure on content safety using the PKU-SafeRLHF dataset, testing the preservation of helpfulness alongside safety leakage.
3. **Bias in Bios (Diagnostic Baseline):** We use gender and profession classification as a replication anchor. This setting provides a linear comparison point to contrast against the nonlinear gains of tangent-constrained erasure.

4.2 Model and Implementation

All experiments are conducted on Qwen2.5-0.5B (24 decoder layers, hidden dimension 896). The current paper-locked comparison set implements five primary erasure methods: linear projection, INLP, LEACE, `gce_ambient`, and `gce_tangent`. LEACE is included as a matched linear baseline in the layer-8 inventory reported below. For `gce_tangent`, we estimate local tangent spaces with PCA-based neighborhoods so that the erasure direction is constrained to a local on-manifold approximation rather than to the full ambient space.

4.3 Evaluation Protocol

All leakage probes are trained from scratch on erased representations to ensure no information leaks from the erasure scorers themselves. We employ three complementary evaluation axes:

Probe-based leakage. A retrained nonlinear MLP probe measures whether concept information remains accessible to a standard function class. We report both linear and nonlinear test accuracy; the gap between them indicates the degree to which concept encoding is nonlinear.

Mutual information via k-NN estimation. Probe accuracy measures whether a *specific function class* can recover the concept; a nonlinear MLP probe might miss information that a more complex architecture could extract. Mutual information $I(\mathbf{X}; y)$ captures *all* shared information between representations and labels, providing a probe-free, architecture-agnostic bound on recoverability.

Mental model. Mutual information decomposes as $I(\mathbf{X}; y) = H(y) - H(y \mid \mathbf{X})$: the entropy of the concept label minus the residual uncertainty after observing the representation. When erasure is effective, $H(y \mid \mathbf{X})$ rises toward $H(y)$ (the label becomes unpredictable from the representation), and MI drops toward zero. For continuous $\mathbf{X}$ and discrete y, we estimate MI using the Ross/Gao k-nearest-neighbor estimator:

$$I(\mathbf{X}; y) = \psi(k) - \big\langle \psi(m_i) \big\rangle + \psi(N) - \big\langle \psi(N_{c(i)}) \big\rangle, \tag{16}$$

where ψ is the digamma function and the procedure is:

1. For each point $\mathbf{x}_i$, find its k-th nearest neighbor *within its class* $c(i)$; call that distance ε_i.
2. Count m_i: the number of *all* points (any class) within distance ε_i of $\mathbf{x}_i$.
3. $N_{c(i)}$ is the total number of points sharing the same label as point i; N is the total sample size.

Intuition. When concept information is encoded in $\mathbf{X}$, same-class points cluster together: ε_i is small, $m_i \approx k$ (mostly same-class neighbors within that radius), and MI is high. After effective erasure, class labels become geometrically mixed: to reach k same-class neighbors one must search a large ε_i, sweeping in many cross-class points so that $m_i \gg k$, and MI drops.

Adaptive resolution and local geometry. A key property of the k-NN estimator is that it sets its measurement scale *adaptively*: for each point, the radius ε_i is determined by the local data density, not by a global bandwidth or histogram bin width. In dense regions of the manifold, ε_i is small; in sparse regions, it is large. The digamma terms $\psi(m_i)$ implicitly encode the local density ratio between the class-conditional and marginal distributions without ever estimating densities explicitly in $\mathbb{R}^d$—the estimator works entirely with neighbor counts, which are topological quantities that naturally handle varying density and curvature.

This makes the estimator well-suited for data on a low-dimensional manifold embedded in $\mathbb{R}^d$. A parametric density estimator (e.g., Gaussian kernel) operating in the ambient space would be misled by the high-dimensional volume: two points close in $\mathbb{R}^d$ may be far apart along the manifold, and vice versa. The k-NN estimator sidesteps this because it only examines local neighborhoods—points that are actually nearby along the data distribution. While it still uses ambient Euclidean distance (not geodesic distance), the locality of k-NN queries means that for small k relative to the dataset, the Euclidean and geodesic neighborhoods approximately coincide.

Connection to TanGCE. Both the MI estimator and TanGCE operate through local neighborhood structure rather than global ambient-space assumptions. TanGCE explicitly projects onto the tangent space estimated from k-nearest neighbors; the MI estimator implicitly adapts to the manifold by measuring information through local ball neighborhoods. Neither assumes the data fills $\mathbb{R}^d$ uniformly. This shared reliance on local structure creates a coherent geometric framework: the same manifold properties that make tangent-projected erasure more precise also make k-NN MI estimation more faithful than ambient-space alternatives.

We compute MI for both concept labels (should decrease after erasure) and control labels (should remain unchanged), yielding a complete picture of erasure selectivity.

Unsupervised SAE diagnostics. We train TopK sparse autoencoders (k=64) on a general-purpose WikiText corpus—independent of any concept labels—then measure concept feature removal rate, collateral damage rate, and feature redistribution score (Section 5). These metrics require no control-task annotations and detect failure modes invisible to probe-based evaluation, such as concept information being redistributed across features rather than removed.

4.4 Geometric Foundation (Appendix Support)

While the main text focuses on the performance of `gce_tangent`, we also provide geometric diagnostics to ground the mechanism story. These include broad geometry sweeps across multiple datasets and layers, subspace projection sensitivity (Q1), perturbation-based subspace sensitivity tests (Q2), and representation connectivity analyses (Q3). These appendix-scale materials support the low-dimensional and nonlinear-geometry motivation for tangent projection, but they should not be read as a standalone causal proof that off-manifold displacement fully explains the observed utility tradeoffs.

5 Results

We evaluate seven erasure methods across three concept domains—sycophancy, gender (Bias in Bios), and safety—at layer 8 of Qwen2.5-0.5B. All methods except TanGCE use λ=0.5; TanGCE uses λ=8.0 to compensate for the naturally attenuated projection norms (see Section 5.4). We measure concept leakage with a retrained nonlinear MLP probe, mutual information via the Ross/Gao k-NN estimator in the original representation space, and unsupervised SAE-based metrics: concept feature removal rate, collateral damage rate, and feature redistribution score.

Table 2: Full erasure method comparison at layer 8 across three concept domains. All methods use λ=0.5 except TanGCE which uses λ=8.0 to compensate for attenuated projection norms (Section 5.4). NL Leak: retrained nonlinear MLP probe accuracy (lower = better concept removal). ΔCtrl: downstream control task accuracy change relative to clean representations (closer to zero = more surgical). Redist%: fraction of non-concept SAE features that become concept-correlated post-erasure (lower = less information reshuffling). Model: Qwen2.5-0.5B. SAE: 8$\times$ TopK, k=64. Control tasks: answer matching (sycophancy), profession classification (gender), helpfulness (safety).

	Sycophancy				Gender (Bias in Bios)				Safety			
Method	NL↓	ΔCtrl	Dmg	Rdst	NL↓	ΔCtrl	Dmg	Rdst	NL↓	ΔCtrl	Dmg	Rdst
Random Global	.748	.000	2.8	0.0	.758	.000	21.9	0.0	.805	+.003	8.2	0.0
Random Per-Samp.	.745	.000	1.4	0.0	.743	+.003	27.3	0.0	.799	−.010	0.0	0.0
Linear Proj.	.740	.000	14.1	0.0	.747	+.001	23.4	0.0	.800	−.003	0.0	0.0
INLP	.734	.000	14.1	0.0	.737	+.001	22.8	0.0	.795	−.004	0.0	0.0
LEACE	.681	.000	1.4	0.0	.739	−.002	29.2	0.0	.785	−.015	4.1	0.0
Ambient GCE	.703	.000	2.8	1.4	.701	.000	15.9	0.0	.778	−.040	0.0	2.0
TanGCE	**.602**	−.015	60.6	**1.4**	**.618**	−.008	51.1	**0.0**	**.663**	+.039	65.3	4.1

Table 3: Mutual information and fidelity metrics (layer 8). MI_c: concept mutual information in original representation space via k-NN estimator (lower = better erasure). MI_{ctrl}: control-label mutual information (higher = better preservation of task-relevant information). Clean baselines: sycophancy MI_c=0.189, MI_{ctrl}=1.157; gender MI_c=0.074, MI_{ctrl}=0.903; safety MI_c=0.160, MI_{ctrl}=0.077. L_2: mean ℓ_2 displacement from clean representations.

	Sycophancy				Gender (Bias in Bios)				Safety			
Method	MI_c↓	MI_{ctrl}	L_2	Ctrl	MI_c↓	MI_{ctrl}	L_2	Ctrl	MI_c↓	MI_{ctrl}	L_2	Ctrl
Random Gl.	.191	1.155	0.02	1.000	.075	.902	0.10	.538	.164	.075	0.06	.565
Linear Pr.	.191	1.155	0.11	1.000	.074	.902	0.06	.540	.161	.074	0.01	.560
INLP	.195	1.158	0.11	1.000	.076	.902	0.06	.540	.161	.075	0.01	.559
LEACE	.198	1.159	0.07	1.000	.075	.901	0.20	.537	.127	.070	0.15	.548
Amb. GCE	.184	1.158	0.02	1.000	.077	.902	0.08	.538	.157	.075	0.04	.523
TanGCE	**.113**	1.158	0.86	.985	**.073**	**.904**	0.41	.531	**.145**	.069	0.65	.602

5.1 TanGCE Achieves Lowest Leakage Across All Domains

Table 2 shows that TanGCE achieves the lowest nonlinear leakage across all three domains: 0.602 on sycophancy (vs. 0.681 for LEACE, 0.703 for Ambient GCE), 0.618 on gender (vs. 0.701 for Ambient GCE), and 0.663 on safety (vs. 0.778 for Ambient GCE). Linear methods (Linear Projection, INLP) barely reduce leakage below random baselines, confirming that concept information is predominantly encoded nonlinearly.

5.2 Control Task Preservation

Table 2 reports ΔCtrl, the change in downstream control task accuracy relative to clean representations. For sycophancy (answer matching), TanGCE incurs a small -1.5pp drop; all other methods preserve perfect accuracy. For gender (profession classification), TanGCE drops only -0.8pp (0.531 vs. 0.539 clean). On safety, TanGCE *improves* helpfulness ($+3.9$pp), suggesting that removing safety-concept directions frees capacity for the helpfulness task.

Table 3 complements this with mutual information: TanGCE preserves control MI across all domains (sycophancy: 1.158 vs. 1.157 clean; gender: 0.904 vs. 0.903 clean). On gender, TanGCE is the only method that slightly *increases* control MI, indicating zero collateral damage to task-relevant information.

5.3 Near-Zero Redistribution

TanGCE maintains near-zero redistribution across all concepts: 1.4% on sycophancy, 0.0% on gender, and 4.1% on safety. In contrast, Ambient GCE shows nonzero redistribution even at λ=0.5, and this grows dramatically at higher λ (Section 5.4).

Table 4: Effect of disruption threshold δ on concept removal and damage rates ($64\times$ SAE). At the lenient $\delta{>}2\%$ threshold, many features register as "damaged" from minor perturbation; at $\delta{>}20\%$, only features with substantial activation change count. TanGCE's damage drops to 1–11% at $\delta{>}20\%$, confirming that TanGCE's collateral impact at lenient thresholds reflects superficial activation shifts rather than genuine information disruption.

Sycophancy						
	$\delta{>}2\%$		$\delta{>}10\%$		$\delta{>}20\%$	
Method	Rem%	Dmg%	Rem%	Dmg%	Rem%	Dmg%
Random Global	0	5	0	0	0	0
Linear Projection	0	24	0	5	0	3
INLP	0	24	0	5	0	3
LEACE	0	3	0	0	0	0
Ambient GCE	0	3	0	0	0	0
TanGCE	33	61	0	18	0	11
Gender						
	$\delta{>}2\%$		$\delta{>}10\%$		$\delta{>}20\%$	
Method	Rem%	Dmg%	Rem%	Dmg%	Rem%	Dmg%
Random Global	50	26	0	1	0	0
Linear Projection	0	29	0	0	0	0
INLP	0	28	0	0	0	0
LEACE	100	35	0	1	0	0
Ambient GCE	0	22	0	0	0	0
TanGCE	50	61	0	10	0	1
Safety						
	$\delta{>}2\%$		$\delta{>}10\%$		$\delta{>}20\%$	
Method	Rem%	Dmg%	Rem%	Dmg%	Rem%	Dmg%
Random Global	13	13	0	0	0	0
Linear Projection	0	2	0	0	0	0
INLP	0	2	0	0	0	0
LEACE	4	7	0	0	0	0
Ambient GCE	4	3	0	0	0	0
TanGCE	83	64	26	21	13	7

Damage threshold sensitivity. Table 4 reveals that the high damage rates at $\delta{>}2\%$ are largely an artifact of the lenient threshold: many non-concept features shift by 2–5% in mean activation—a perturbation too small to meaningfully alter downstream behavior (consistent with preserved $\mathrm{MI}_{\mathrm{ctrl}}$ in Table 3). At the strict $\delta{>}20\%$ threshold, TanGCE's damage drops to 1% on gender and 7% on safety, confirming that TanGCE's apparent collateral impact at the lenient threshold reflects superficial activation shifts, not genuine information disruption.

5.4 The λ Sweep: Off-Manifold Displacement Creates Spurious Correlations

Table 5 reveals a critical failure mode of Ambient GCE: at $\lambda \geq 4$, concept MI *increases* rather than decreases (negative MI drop). On sycophancy at $\lambda{=}8$, the MI drop is -0.063; on gender, -0.098; on safety, -0.092. Simultaneously, redistribution reaches 15–31%.

This occurs because ambient erasure pushes representations off the data manifold. The displacement magnitude correlates with each sample's concept signal strength, creating a new axis of variation that SAE features detect as concept-correlated. Concept information is laundered into the geometry of the displacement rather than removed.

TanGCE avoids this failure mode through two mechanisms. First, projection onto the tangent space removes off-manifold gradient components, ensuring displacement stays on the manifold. Second, retaining the raw projection norm $\|\mathbf{P}_T(\nabla f)\|$ as a confidence weight naturally attenuates erasure for samples where the concept gradient is mostly off-manifold. This yields monotonically decreasing leakage with increasing λ, with redistribution remaining below 8% even at $\lambda{=}8$.

Table 5: λ sweep comparing Ambient GCE, TanGCE (renorm), and TanGCE at layer 8. TanGCE retains the raw tangent projection norm as a confidence weight: if the gradient is mostly off-manifold, $\|\mathbf{P}_T(\nabla f)\| \ll 1$ and erasure is naturally attenuated. Ambient GCE at high λ increases MI_c (negative MI drop) and shows high redistribution—concept information is reshuffled into new SAE features rather than removed. TanGCE maintains near-zero redistribution across all λ values.

Concept	Method	λ	NL↓	MI drop↑	Remove%	Dmg%↓	Redist%↓
Syco.	Ambient	0.5	.678	+.007	0	1.4	1.4
	Ambient	8.0	.734	−.063	25	23.3	**15.1**
	TanGCE	4.0	.596	+.006	50	27.4	5.5
	TanGCE	8.0	**.588**	+.014	50	49.3	4.1
Gender	Ambient	0.5	.666	−.001	21	23.4	0.0
	Ambient	8.0	.702	−.098	79	78.8	**11.1**
	TanGCE	4.0	.580	−.000	26	41.5	0.0
	TanGCE	8.0	**.470**	−.002	37	57.6	0.2
Safety	Ambient	0.5	.664	+.008	0	1.9	0.0
	Ambient	8.0	.770	−.092	12	30.8	**30.8**
	TanGCE	4.0	.680	+.015	44	46.2	5.8
	TanGCE	8.0	**.646**	+.011	68	67.3	7.7

Table 6: Effect of SAE dictionary size on concept removal detection. We train TopK SAEs (k=64) at $8\times$ and $64\times$ expansion on the same general-purpose WikiText corpus and re-evaluate all methods. Concept removal rate (%) measures how many concept-correlated features are disrupted by erasure; damage rate (%) measures collateral disruption to non-concept features. Correlation threshold $|r|>0.1$; disruption threshold $\delta>0.02$. The $8\times$ SAE (7,168 features) systematically underestimates concept removal due to polysemantic feature representations, particularly on gender where it reports 0% removal for methods that demonstrably erase the concept.

	Sycophancy				Gender (Bias in Bios)				Safety			
	Remove%		Damage%		Remove%		Damage%		Remove%		Damage%	
Method	$8\times$	$64\times$	$8\times$	$64\times$	$8\times$	$64\times$	$8\times$	$64\times$	$8\times$	$64\times$	$8\times$	$64\times$
Linear Proj.	14	14	14	19	0	0	23	29	0	0	0	2
INLP	14	14	14	19	0	0	23	28	0	0	0	2
LEACE	14	0	1	3	**0**	**100**	29	35	9	5	4	5
Ambient GCE	0	0	3	1	0	0	16	20	0	5	0	2
TanGCE	71	71	61	60	**0**	**100**	51	60	59	82	65	68

5.5 Polysemanticity Confounds Linear Correlation Metrics

A natural approach for measuring concept removal in SAE feature space is to identify concept-correlated features via point-biserial correlation $r(z_j, y)$ and then measure what fraction are disrupted after erasure. However, this approach assumes that each SAE feature encodes a single semantic concept—an assumption that fails when features are *polysemantic*.

Consider a polysemantic feature z_j that activates when *both* concept y=1 and topic t=science are present, or when y=0 and t=politics. The marginal correlation $r(z_j, y) \approx 0$ because the feature fires equally often for both concept labels, yet z_j carries substantial concept information *conditioned on topic*. A linear correlation metric would classify this feature as non-concept-related and miss the fact that erasure correctly disrupts it.

Table 6 demonstrates this effect empirically. At $8\times$ expansion (7,168 dictionary features), the SAE reports **0% concept removal** for both TanGCE and LEACE on gender—despite both methods substantially reducing nonlinear probe leakage (Table 2). Increasing the dictionary to $64\times$ expansion (57,344 features) resolves the polysemanticity: features become more monosemantic, linear correlations faithfully detect concept encoding, and both methods now show **100% concept removal**.

The effect is concept-dependent. Sycophancy removal rates are stable across expansions (71% at both $8\times$ and $64\times$ for TanGCE), suggesting that sycophancy features are already relatively monosemantic at $8\times$. Gender features, by contrast, are heavily polysemantic at $8\times$—the concept is distributed across

Table 7: Geometry contrast between sycophancy and gender across completed layers 4, 8, and 12. Lower intrinsic dimension and larger local-vs-global angle indicate stronger nonlinearity: a concept whose erasure direction varies significantly across the input space, making a single fixed direction insufficient to capture the full concept. Sycophancy shows substantially higher nonlinearity (dim 34–53, angle 76.9–82.6°) compared to gender (dim 365–461, angle 51.8–55.8°), supporting sycophancy as the primary case for manifold-aware erasure methods. This table demonstrates the geometric contrast but does not establish causality: it does not prove that the observed nonlinearity will be converted into lower displacement or higher utility preservation by any particular erasure method.

Concept	Local intrinsic dimension	Mean local/global angle
Sycophancy	34–53	76.9–82.6°
Gender	365–461	51.8–55.8°

features that jointly encode gender with other attributes, making it invisible to marginal correlation. Safety shows an intermediate pattern: removal increases from 59% to 82%.

These results have two implications. First, SAE-based erasure evaluation requires sufficiently high dictionary expansion to overcome polysemanticity; $8\times$ expansion is inadequate for concepts that are entangled with other attributes. Second, the zero-redistribution result for TanGCE on gender (Table 2) is robust: redistribution remains at 0% even at $64\times$ expansion, confirming that TanGCE removes rather than reshuffles concept information.

5.6 Geometric Evidence and Off-Manifold Displacement

Table 7 demonstrates that sycophancy is much more nonlinear than gender: its intrinsic dimension (34–53 across layers 4, 8, 12) is far lower than gender's (365–461), and its local-vs-global angle is much larger (76.9–82.6° vs. 51.8–55.8°). This geometry contrast validates sycophancy as the primary case for manifold-aware erasure methods, while the strong TanGCE results on gender demonstrate that the method generalizes even to approximately linear concepts.

Appendix-Supporting Evidence. Detailed geometric analyses supporting these claims are provided in the Appendix, including:

- **Subspace Sufficiency (Q1):** Evidence that task-specific computation is concentrated in a low-dimensional linear subspace.
- **Perturbation Sensitivity (Q2):** Comparison of perturbations in high-variance and orthogonal subspaces.
- **Representation Connectivity (Q3):** Analysis of the connected, approximately convex topology of the representation manifold.
- **Multi-Layer Accumulation:** Results on how projection error accumulates across the network.

6 Discussion

The completed paper suite already narrows the central claim. In every finished comparison, the tangent-based method removes more measured concept signal than its ambient counterpart. That pattern appears in the sycophancy sweeps, in the completed layer-8 safety sweep, and in the Bias in Bios diagnostic. What the current paper snapshot does not yet show is the cleaner second half of the original thesis in which tangent projection also achieves lower displacement or uniformly better utility retention. In the completed runs, tangent reaches lower leakage by taking larger steps, and its utility metrics are either slightly worse or mixed.

The geometry diagnostics still matter because they explain why sycophancy is the harder case. Sycophancy shows local-vs-global angles near $80°$, while gender remains near $53°$, so a single global erasure direction is a much worse summary for the primary nonlinear concept. That is consistent with tangent-based methods improving removal in both concepts while leaving sycophancy as the clearly

more nonlinear case, but the finished results currently frame tangent as a stronger remover rather than a dominantly better Pareto point.

The appendix snapshot tables slightly sharpen the near-term agenda. In the broader repository reports (Tables 8 and 9), the same removal advantage continues at layers 16 and 23 for both sycophancy and safety, although the utility metrics differ by domain and report. We do not elevate those rows to headline claims because they were not part of the paper-locked evidence set, but they strengthen the case for spending the next round of effort on matched-leakage comparisons and deeper-layer replications rather than on searching for a different primary mechanism.

Two implications follow. First, the paper should presently avoid claiming that off-manifold control is already proven to improve the full leakage-utility frontier; the matched-leakage analysis and broader replications are still needed for that claim. Second, the current evidence suggests that the next important question is not whether tangent projection changes the erasure direction at all, but under which layers and operating points that removal advantage can be converted into a cleaner utility tradeoff.

The matched layer-8 linear-family comparison clarifies one part of that tradeoff. LEACE is more surgical than tangent on this slice because it removes linearly decodable concept signal while taking a much smaller geometric step and preserving more downstream utility, especially on sycophancy where the answer-matching penalty is substantially smaller. That does not make LEACE a substitute for tangent on the main nonlinear claim: LEACE succeeds by solving the linear erasure problem cleanly, whereas tangent remains the stronger method when the target concept is evaluated with a fresh nonlinear probe. The current evidence therefore suggests a division of labor rather than a single winner: LEACE is the cleaner linear baseline, while tangent remains the more aggressive nonlinear remover.

7 Limitations

The current manuscript deliberately limits its main-text quantitative claims to the paper runs that were complete when this draft was assembled. That keeps the claims auditable, but it also means the strongest version of the utility-preservation story is not yet established in the locked evidence set. The deeper iterative sycophancy and safety reports are directionally consistent with the main narrative and are included in Appendix A, yet the paper's central claims still rely on the smaller completed suite.

A second limitation is measurement scope. All experiments use Qwen2.5-0.5B, and utility is measured through task-specific proxies such as answer matching, helpfulness ranking, profession classification, and dataset-source retention. These metrics are appropriate for surgicality, but they do not yet establish broad capability preservation across unrelated tasks.

Finally, the geometry diagnostics support low-dimensional structure, concept nonlinearity, and connectedness, but they do not by themselves prove that curved-manifold effects are the sole causal mechanism behind every leakage–utility tradeoff. The current paper therefore treats geometry as strong supporting evidence for tangent projection rather than as a complete causal proof.

8 Conclusion

The completed paper runs already support a narrower, evidence-backed conclusion. Sycophancy is geometrically more nonlinear than Bias in Bios, and tangent-based erasure consistently achieves lower measured concept leakage than the completed ambient baselines on sycophancy, safety, and gender. That makes manifold-aware erasure a supported hypothesis for stronger concept removal in nonlinear settings.

At the same time, the current draft should stop short of claiming that tangent erasure already dominates the utility-fidelity tradeoff. In the finished runs, the lower-leakage tangent points typically require larger displacement and often incur equal or larger utility loss than ambient baselines. Appendix snapshot tables are directionally consistent with the same removal advantage at deeper layers, but the paper is strongest if it uses them as status evidence rather than as the basis for stronger causal or Pareto-frontier claims. The manuscript therefore lands best as support for the geometry contrast and for tangent's removal advantage, while reserving the broader utility-preservation claim for matched-leakage analyses and future replications.

Table 8: Repository snapshot from the iterative sycophancy report at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is answer-matching delta relative to clean representations. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This snapshot table is not a main-text claim anchor and should be read only as evidence of directional consistency with the layer-8 paper-locked row (Table 2). Deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines across this exploratory sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.063	0.213	-20.7pp	-0.5pp	2.88	0.62
16	0.153	0.400	+0.0pp	+0.0pp	3.79	0.27
23	0.138	0.361	-0.1pp	-0.2pp	44.50	3.11

Table 9: Repository snapshot from the iterative safety report at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is pairwise helpfulness delta. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This snapshot table is not a main-text claim anchor and should be read only as evidence of directional consistency with the layer-8 paper-locked row (Table 2). Deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines across this exploratory sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.0725	0.2583	-0.052	-0.023	2.70	0.18
16	0.1292	0.3517	-0.092	+0.003	4.37	0.60
23	0.1483	0.3404	-0.062	-0.152	38.32	5.30

A Repository Snapshot Tables

This appendix records broader repository snapshot tables that are directionally consistent with the main text but were not part of the locked paper-evidence set when the quantitative claims in Section 5 were written. We include them to document the current state of the project and to guide the next round of matched-leakage and deeper-layer analyses. The central paper claims continue to rely only on the completed paper suite summarized in the main text.

A.1 Iterative Sycophancy Snapshot

A.2 Iterative Safety Snapshot

Geometry Note. The broader geometry report continues to support three useful points: early and mid-layer representations are effectively low-dimensional, on-manifold directions carry much of the model's sensitivity in these perturbation tests, and interpolation remains consistent with a connected representation space. At the same time, the Q2 perturbation study is best read as evidence of sensitivity concentration in a high-variance subspace rather than as a standalone proof of curved-manifold causality.

Table 10: Geometry evidence map organizing the appendix diagnostics into four research blocks. Each block addresses a distinct aspect of manifold structure: Q1 examines whether task-relevant computation concentrates in a low-dimensional linear subspace, Q2 tests whether perturbations on the manifold are more consequential than off-manifold perturbations, Q3 validates whether the manifold is connected across domains, and multi-layer accumulation tracks error compounding through the network. These diagnostics support the mechanism discussion and motivate the need for manifold-aware erasure methods but do not by themselves establish causality.

Experiment Block	Research Question	Primary Evidence
Q1: Rank-retention	Subspace concentration	Top-1 agreement under intrinsic projection
Q2: Perturbation sensitivity	High-variance vs. orthogonal directions	KL and top-1 under on/off-subspace noise
Q3: Domain connectivity	Shared vs. domain manifolds	Cross-domain interpolation smoothness
Multi-layer accumulation	Error compounding across the network	Simultaneous intrinsic projection across layers

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.

- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not

including code, unless this is central to the contribution (e.g., for a new open-source benchmark).

- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.

- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.

- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.